

Skye Lane Goetz

Applied AI/ML Engineer || Biomedical Data Scientist

Work Experience

Institute for Systems Biology (ISB)

Research Associate (Data Science), Hood Lab

Seattle, WA

Nov 2024 – Present

- Engineered MultiomicsKG, a 1M+ edge multiomics knowledge graph integrating data across 250+ papers for the NCATS Biomedical Data Translator (superset of published MicrobiomeKG); trained an RGAE in PyTorch Geometric on SapBERT embeddings for link prediction (Hits@10 = 0.47), with GPU-accelerated HDBSCAN/UMAP clustering via cuML
- Designed [Tablassist](#), a polyglot AI-orchestration platform (7 internal users) dual-published to PyPI (Python CLI, 30+ commands) and npm (TypeScript OpenCode plugin); wires a 5-agent system (Configurator/Pioneer/Builder/Scout/Extractor) with ledger-backed autonomous PMC discovery surviving context resets, validation-as-a-hook on YAML writes, and ground-truth Biolink schema injection
- Refactored knowledge-graph extraction in [Tablassert](#) (PyPI + GHCR, v7.4.10, Production/Stable) from 800GB SQLite to 250GB DuckDB OLAP while cutting query time from ~10 min to 4–11 s (~100–1000×) using Polars lazy frames, Pydantic v2 strict models (170+ Biolink categories), and 3-stage entity-resolution QC (exact → RapidFuzz → BioBERT/ONNX)
- Built Go infrastructure for biomedical entity resolution: [Datassert](#), a 4-stage ETL sharding BABEL exports across 10 indexed DuckDBs via xxhash64 with cache-line-padded mutexes and bounded errgroup parallelism; and a [Gin API](#) exposing CURIE-resolution and PMC-streaming endpoints with custom Redis auth (SHA-256 → xxhash64) over a 15 TB PMC mirror
- Co-mentored a master's student on a [clinical trial classifier](#), extending the pipeline with BERT sentence transformers, incremental PCA, Autoencoders, and LightGBM to lift AUC from 0.63 to 0.90 across 3 datasets
- Published [MicrobiomeKG](#), a 115k-edge microbiome KG spanning 40 curated papers with ResNet-based edge scoring

OpenAI-Sponsored Poly Prompt Hackathon (Cal Poly)

AI/Tech Lead – Parker: The KG Study Tool

San Luis Obispo, CA

Feb 2026 – Feb 2026

- Led a [LangGraph pipeline](#) for course-material knowledge graph generation using Canvas API ingestion, [LinkML ontology modeling](#), DuckDB storage, and LLM entity extraction from PDFs, PPTs, and Excel files deployed using AWS EC2

Publications

- Skye L. Goetz**, Amy K. Glen, and Gwênlyn Glusman. MicrobiomeKG: Bridging Microbiome Research and Host Health Through Knowledge Graphs. *Frontiers in Systems Biology*, 2025. doi:10.1101/2024.10.10.617697.
- Karamarie Fecho et al. Announcing the Biomedical Data Translator: Initial Public Release. *Clinical and Translational Science*, 2025. doi:10.1111/cts.70284.

Skills

Languages: Python, Go, Elixir, TypeScript, R, SQL, Shell, LaTeX

AI/ML: PyTorch, LightGBM, Scikit-learn, LangGraph, MCP, Transformers, PEFT, Diffusers, Ollama, Snorkel

Data & Storage: Polars, DuckDB, PostgreSQL, Redis, SQLite, ChromaDB, NumPy

Knowledge Graphs: GNNs, PyTorch Geometric, Ontology NER, Neo4j, KGX, LinkML, NetworkX, Centrality Algorithms

NLP: spaCy, Autoencoders, Sentence Transformers, BERT, Text Classification, PCA, KernelPCA

DevOps/Infrastructure: AWS, Nix/NixOS, Docker, QEMU/KVM, FastAPI, Gin, GitHub Packages/Actions

Domain: Multi-Omics, Clinical Trials, Polygenic Risk Scores, Quantum Chemistry, TD-DFT

Education

California State Polytechnic University, San Luis Obispo

B.S. Biochemistry (GPA: 3.626/4.000)

San Luis Obispo, CA

Feb 2024 – Jun 2026 (Expected)

- Applied ML consulting on senior capstone projects (+0.15 AUC improvement)
- Developed ML pipeline for biochemical assay prediction (+0.08 R²)